

X58/P45/P43/P41 series of motherboards from GIGABYTE will now be equipped with Creative's Sound Blaster X-Fi technologies

NVIDIA's Ion platform uses the Intel Atom and their own GeForce 9400 graphic solution

TRIED & TESTED

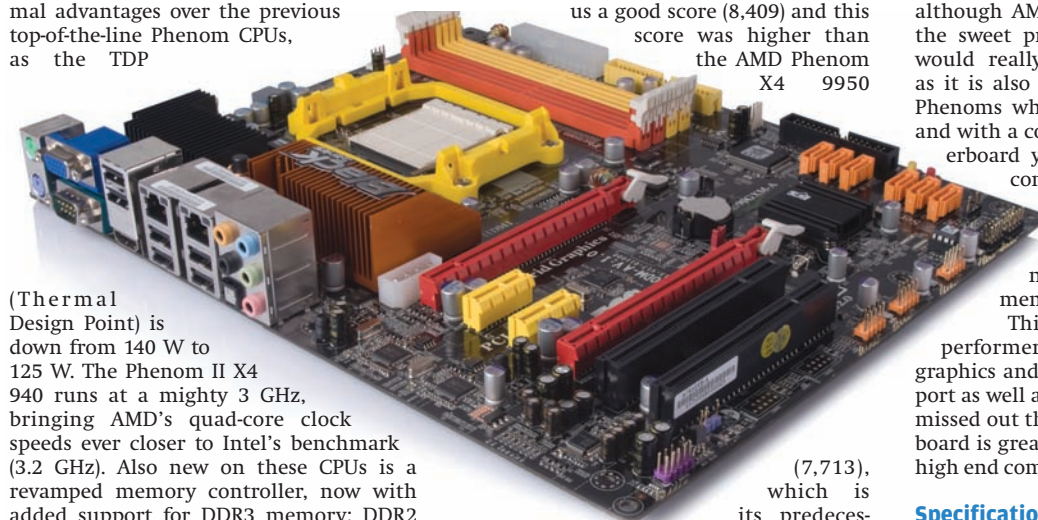
AMD Phenom II X4 940 and ECS A790GXM-A

Still Fighting On

The AMD X4 940 is the top-of-the-line Phenom II processor and the latest from AMD's stables. This CPU is based on its shiny new 45 nm fabrication process and represents a step up in terms of performance from its Phenom processor range. The Phenom II brings thermal advantages over the previous top-of-the-line Phenom CPUs, as the TDP

We used 2 x 2GB of Corsair Dominator memory (1,066 MHz) and an ASUS GTX285 TOP for the test. The ECS motherboard was also tested separately without the graphics card in order also to test its integrated graphics solution.

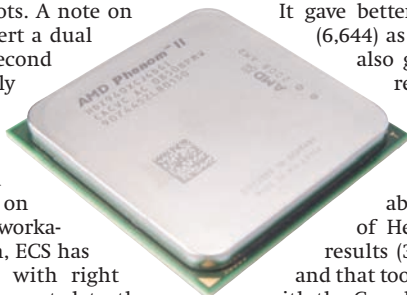
In PC Mark 05 the processor gave us a good score (8,409) and this score was higher than the AMD Phenom X4 9950



(Thermal Design Point) is down from 140 W to 125 W. The Phenom II X4 940 runs at a mighty 3 GHz, bringing AMD's quad-core clock speeds ever closer to Intel's benchmark (3.2 GHz). Also new on these CPUs is a revamped memory controller, now with added support for DDR3 memory; DDR2 support is still retained, therefore AMD has a compatibility advantage when compared to Intel's Nehalem. Another step up from the Phenom CPUs lies in the additional L3 cache which goes up from 2 MB to 6 MB. For reference the fastest Phenom processor in terms of clock speed was the X4 9950 clocked at 2.6 GHz.

The ECS A790GXM-A motherboard is part of its *Black Series* and is based on the 790GX chipset from AMD. The Northbridge, which is the AMD 790GX, is covered by a neat-looking copper heatsink while the Southbridge, the AMD SB750 is covered by a black aluminium heatsink. There is another black aluminium heatsink present above the Northbridge, which takes care of cooling the MOSFETs around the CPU region.

This board also comes with two PCIe x16 slots and supports CrossFire. With a single card, obviously the slot runs at x16 speeds. There are also two PCI slots as well as two PCIe x1 slots. A note on the layout – if you insert a dual slot card into the second PCIe x16 slot it not only blocks a PCI slot but also the first two SATA ports as well as the small power and reset buttons present on the board. As a partial workaround for this problem, ECS has supplied SATA cables with right angled heads to be connected to the SATA ports.



(7,713), which is its predecessor. 3D Mark 06 showed us the same result in a higher score (3,958), but it was pretty close as the Phenom II was barely able to beat the X4 9950 which gave us 3,886. In Cinebench R10 the processor gave us a score of 9,555, which again did much better than the earlier Phenom with a score of 8,026. Similar results were obtained with PovRay 3.7, Maya 8.0, WinRAR 3.8, DivX 6.8.0.19. In Doom 3 we obtained a score of 203.2 FPS as compared to 196 FPS from the X4 9950. The Phenom II was not able to beat the X4 9950 when it came to Company of Heroes, though. In mostly all of the benchmarks the Phenom II X4 940 outperformed the Phenom X4 9950.

When it comes to the ECS motherboard, in comparison to our last motherboard test (with integrated graphics), the ASUS Crosshair II Formula (based on a GeForce 780a SLI chipset) was the best performer, but this board did much better.

It gave better scores in PC Mark 05 (6,644) as well as SiSoft Sandra. It also gave us better results in real world DivX encoding as well as file transfer tests. Doom 3 at ultra settings and 1024x768 resolution gave us playable FPS (36.1). Company of Heroes gave us great FPS results (32.7) at 800x600 as well, and that too more than was obtained with the Crosshair II Formula.

This AMD Phenom II 940 is definitely

better than the old line of Phenom processors, but when you put them in comparison to a higher end Intel Core 2 Quad such as the Q9550 it can't compete. Therefore clock to clock AMD's architecture still falls behind Intel's Core architecture. The new Intel Nehalem CPUs are faster still, although AMD's main attraction remains the sweet pricing: Rs. 13,750. AMD fans would really appreciate the Phenom II as it is also more stable than the earlier Phenoms when it comes to overclocking, and with a combination of an AM2+ motherboard you would not need to do a complete overhaul of your PC, as you can retain the DDR2 RAM; this is definitely a plus considering the astronomical prices of good DDR3 memory.

This motherboard is a good performer for one with integrated graphics and it even comes with an HDMI port as well as the usual D-Sub port (DVI is missed out though). Priced at Rs. 7,875 this board is great for those wanting to build a high end computer with an AMD processor.

Specifications (Phenom II X4 940):

Clock speed: 3.0 GHz, HyperTransport Link: 3.6 GT/s, Socket: AM2+, TDP: 125 W, Total L2 cache: 2 MB, Total L3 cache: 6 MB

Specifications (ECS A790GXM-A):

Socket AM2+, North Bridge: AMD 790GX, South Bridge: AMD SB750, Graphics: AMD 790GX-based with ATI Radeon HD3300 graphics, 2 x PCI Express x16 slots, 2 x PCI Express x1 slots, 2 x PCI slots

Ratings for Phenom II X4 940

Features: 8.5
Performance: 8
Value for money: 8
Overall: 8

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Price: Rs. 13,750

Ratings for ECS A790GXM-A

Ratings
Features: 7.5
Performance: 8
Build quality: 7
Value for money: 8
Overall: 7.5

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